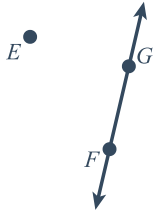


Review: Points, lines, and planes

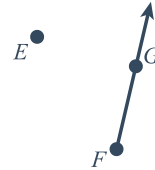


1 Name the line, line segment, or ray in each diagram below using geometric notation.

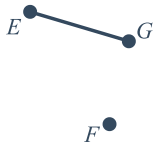
a



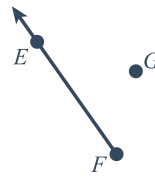
b



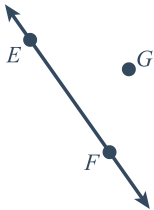
c



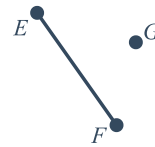
d



e

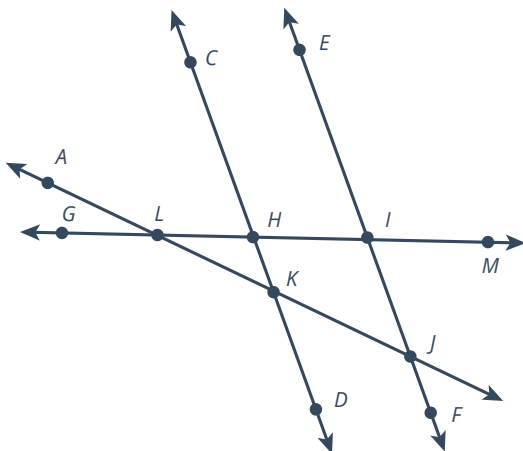


f

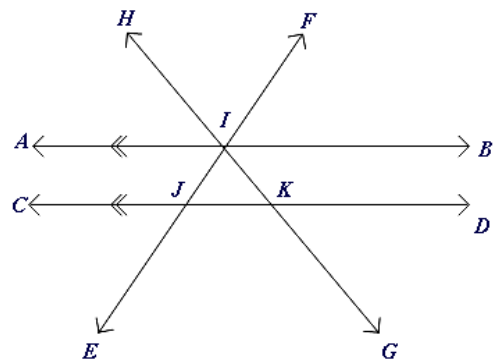


2 Name three different pairs of collinear points in each diagram below.

a

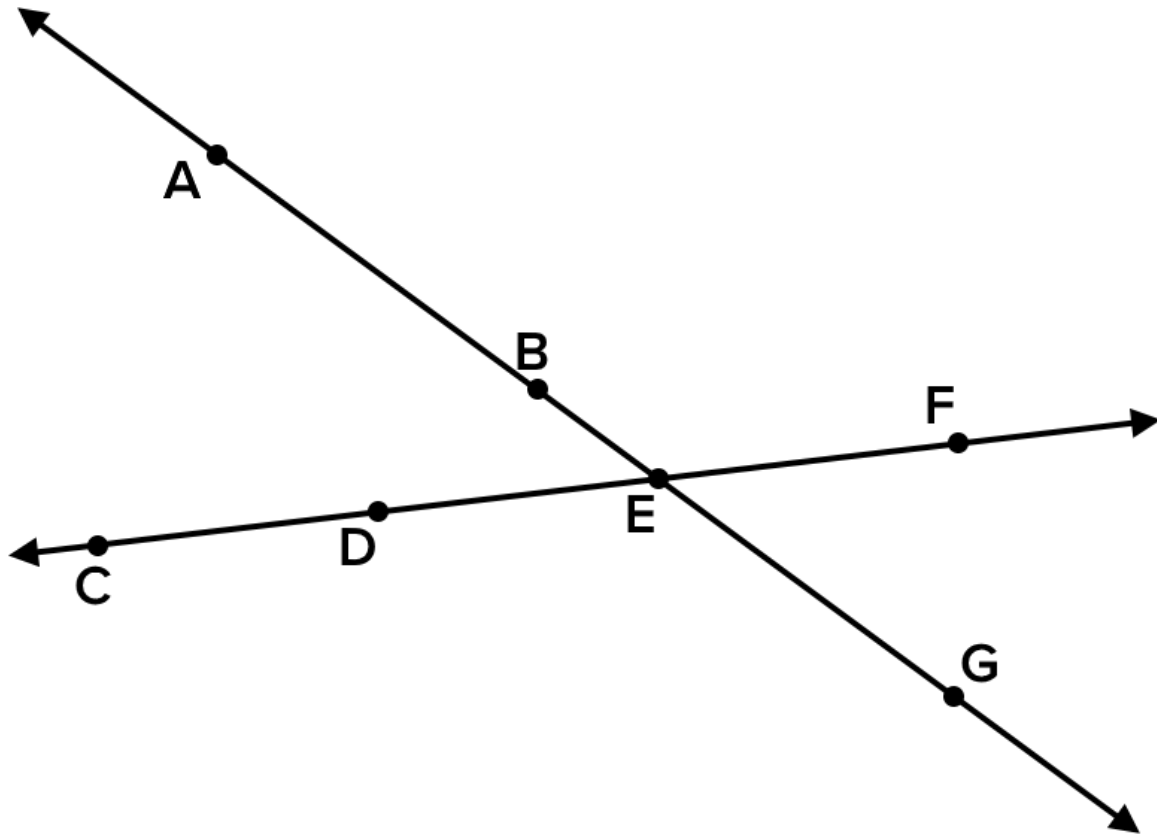


b

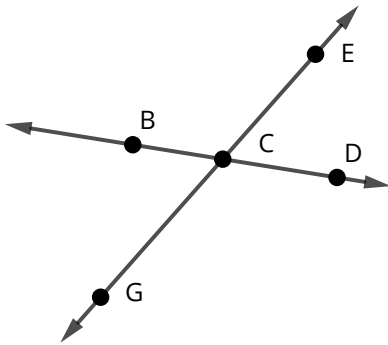


3 What is another way to define the line $\overleftrightarrow{CD}$?

4 What is another way to describe the same set of points as $\overleftrightarrow{FD}$?



5 Determine whether the statement is true or false based on the diagram.

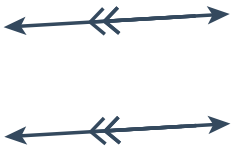


- a Point C is on $\overleftrightarrow{GE}$.
- b Point C is on $\overleftrightarrow{BD}$.
- c The intersection of $\overleftrightarrow{GE}$ and $\overleftrightarrow{BD}$ is point C .
- d $\overleftrightarrow{GE}$ and $\overleftrightarrow{BD}$ must be perpendicular because they intersect.

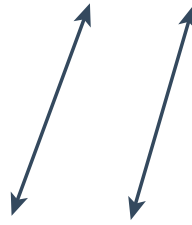


6 In each diagram below, state whether the lines are parallel or not parallel.

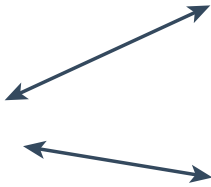
a



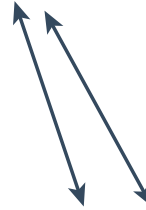
b



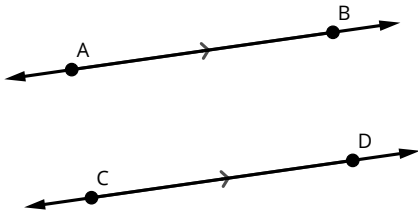
c



d



7 Consider the diagram below.



State whether each of the following statements is true, false, or if there is not enough information to decide.

a $\overline{AB} \parallel \overline{CD}$

b $\overline{AB} \cong \overline{CD}$

c $\overrightarrow{AB} \parallel \overrightarrow{CD}$

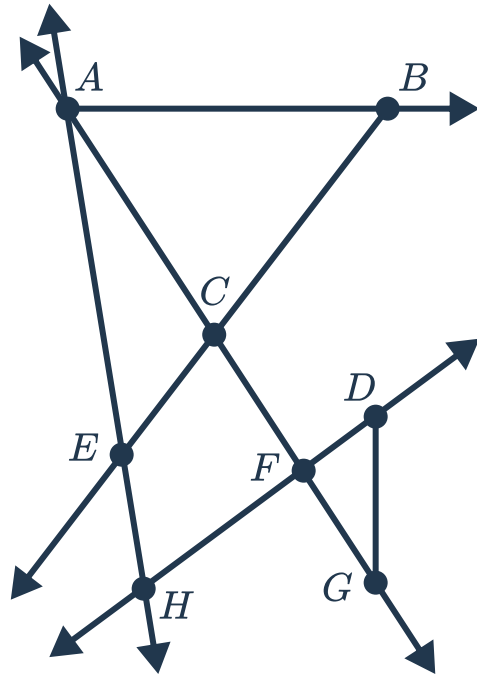
d $\overrightarrow{AB} \parallel \overrightarrow{CD}$



Additional questions

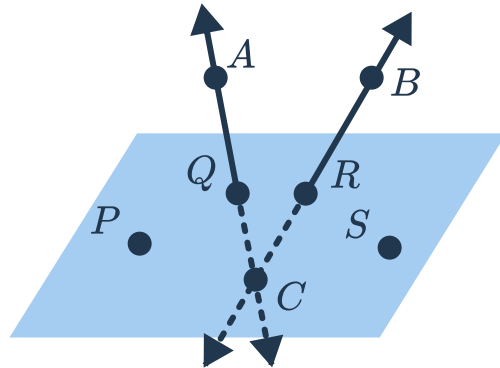
8 Using proper notation, identify the following features of the diagram shown:

- a A set of three collinear points
- b A line passing through point F
- c A ray starting from point A
- d Another name for the ray $\overrightarrow{BE}$

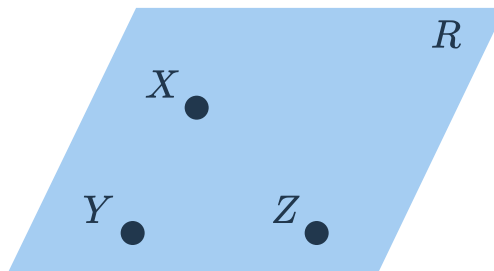


9 Using proper notation, identify the following features of the diagram shown:

- a A line that is contained in plane PQR
- b A line that intersects, but is not contained in, plane PQR
- c Four coplanar points
- d Three collinear points
- e An angle that is contained in the plane
- f An angle that is not contained in the plane
- g A pair of opposite rays

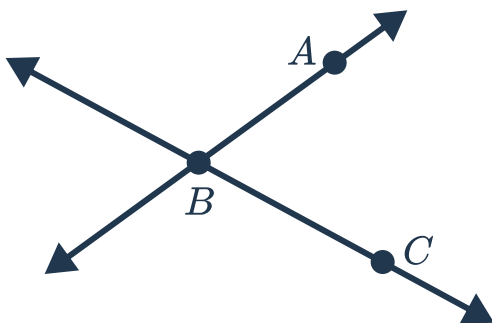


10 Name the geometric figure shown below in different ways:

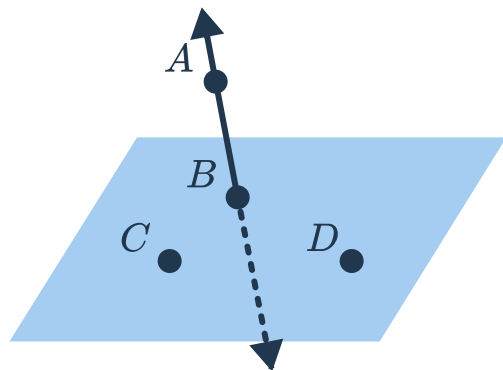


11 State the intersection of the two geometric figures shown in each of the following diagrams:

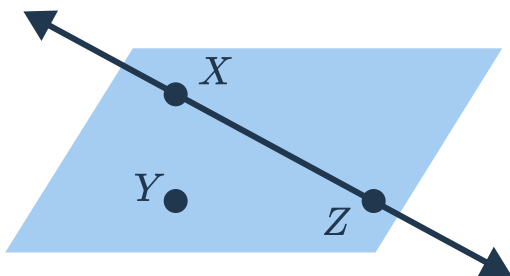
a Lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{BC}$



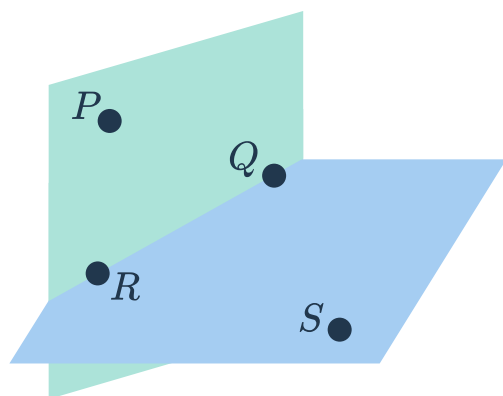
b Line $\overleftrightarrow{AB}$ and Plane BCD



c Line $\overleftrightarrow{XZ}$ and Plane XYZ



d Planes PQR and QRS



12 Answer the following questions:



- a Draw a ray that has point A as an endpoint.
- b Write a name for the ray you have drawn.
- c How many points did you need to use to name the ray?
- d When naming the ray, does the order of the points matter? Explain your answer.

13 Lines $\overleftrightarrow{PQ}$ and $\overleftrightarrow{RS}$ meet at point M .

- a Draw a diagram that shows this.
- b Identify a pair of opposite rays in the diagram.

14 Planes DEF and EFG have common points of E and F .

- a Draw a diagram that shows this.
- b Name the full intersection of the two planes.

15 Draw a diagram that shows $\overline{AB}$ contained in plane P , with $\overleftrightarrow{CB}$ intersecting (but not contained in) the plane.

16 Determine whether each of the following statements are true or false. If true, draw an example. If false, draw a counter example.

- a Through any two points, there is exactly one line.
- b Two distinct pairs of points always form two different lines.
- c If two points are both contained in a plane, then the line that they form is also entirely contained in that plane.
- d For any three collinear points, there is exactly one plane that contains them.
- e If three points are collinear, they cannot be contained in a single plane.
- f For any two distinct lines, there is a plane which contains them both.

17 For each of the following geometric figures, determine a real life example which is a model for the figure. Are there any limitations or restrictions on the real life example?

- a A ray
- b A line
- c A plane
- d Skew lines